

Optimal carbon tax with dynamic skill heterogeneity

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1 Introduction

How can we design fair carbon taxation in the context of vast inequalities between households? Is the carbon tax too high, too low, or just right? Should everyone be expected to pay the same price on their emissions regardless of their income? How do we address the fact that some individuals may be more vulnerable to the impacts of climate change than others? There is growing interest in policy circles in answering these questions, as evidenced for example by the many references to the need for climate policy to consider within-country inequalities in addition to between-country inequalities in the sixth IPCC assessment report (IPCC 2022).

Integrated assessment models, which are macroeconomic growth models with climate modules commonly used to estimate the optimal level of a carbon tax, often view each country or region as a single unit. As such, all within-country inequalities with regards to income and climate impacts are abstracted away when determining the best level of the carbon tax. Up until now, most research on the distributional impact of carbon taxation takes an *ex post* perspective, asking whether existing carbon taxes are regressive and comparing the progressivity of different revenue recycling schemes. However, there is as of yet little practical guidance on if and how income and climate impact inequalities may affect the optimal design of the carbon tax itself. In this paper, I analyze the joint optimal design of labor income taxes, capital income taxes, and carbon taxes in a dynamic Mirrlees model with pollution and heterogeneous climate damages where higher skill agents are less affected by climate change. This follows recent evidence that individuals with lower income and worse health are more vulnerable to climate change (CITE).

While some theoretical results have been derived in static models with income inequality (Cremer et al. 1998; Jacobs and de Mooij 2015), they cannot be directly applied to integrated assessment models which are dynamic in nature. Since such models are used by policymakers to quantify the appropriate level of a carbon tax (Environment and Climate Change Canada 2023; Interagency Working Group on Social Cost of Greenhouse Gases 2021), this is an important first step towards bridging theory and practice.

This research is part of the literature on second-best carbon taxation. A number of studies have shown that introducing an exogenous revenue requirement financed with distortionary income taxation to a representative agent model with a climate externality lowers the optimal level of the carbon tax because environmental quality competes with revenue collection (Goulder 1995; Bovenberg and de Mooij 1994; Barrage 2019). By calibrating a dynamic Ramsey taxation model augmented with a climate component to fiscal data from 107 countries, Barrage estimates the optimal carbon tax to be 8-24% lower than the values estimated in a first-best environment. Such a result may partially explain why the price of carbon in most countries falls short of the value recommended by standard climate-economy models. That is, countries may be reluctant to implement bolder climate policies because they worry about their ability to raise revenue in addition to the usual political-economy reasons.

However, using a static Mirrlees taxation model enhanced with a climate component, Jacobs and de Mooij (2015) show that a tax on polluting goods needs not be adjusted for the presence of income taxation if utility is weakly separable and the government is free to levy a non-linear labor income tax. Otherwise, if the polluting good is relatively more complementary with leisure, if the polluting good is relatively more demanded by high-ability households, or if the government cannot redistribute as much as desired, the optimal tax on polluting goods may be lower or higher than in a first-best environment. Hence, higher carbon taxes do not necessarily contradict the government's fiscal goals as implied by a representative agent model.

Douenne et al. (2023) extend the representative-agent Ramsey setting in Barrage 2019 to a heterogeneous-agent Ramsey setting using a methodology developed in Werning 2007 to explore how inequality affects the structure of the optimal carbon tax. Similarly to Barrage, they find that the part of the carbon tax correcting for utility damages of pollution should be divided by the marginal cost of public funds while the part correcting for output damages remains unaffected. Using a model calibrated to US data, they find that the social cost of carbon is 3.9% lower than in representative-agent models. However, their approach assumes that markets are complete such that agents can perfectly insure themselves against idiosyncratic income shocks * and limits the planner's tax instruments to be linear.

WRITE ALL OF THIS IN A MORE COHERENT MANNER. JUSTIFY.

*(WHY DOES THAT MATTER? HOW DOES THAT AFFECT THE ARGUMENT?)

2 Contribution

3 Environment

I consider a dynamic Mirrlees model of income taxation with two goods: one "clean" good c and one "polluting" good q . Consumption of the polluting good generates a pollutant stock E according to a function of last period's pollutant stock and the current total consumption of polluting goods. This characterization of the evolution of the pollutant stock allows for current emissions to affect future pollutant accumulation, thus possibly affecting future welfare.

$$E_t = S \left(E_0, E_1, \dots, E_{t-1}, \int q \, dF \right)$$

There are T periods denoted by $t = 0, \dots, T$ where T may potentially be infinite and preferences are time separable but otherwise allow arbitrary interactions between current clean goods consumption c , current polluting goods consumption q , labor l , and pollution disutility h . Pollution disutility is a function of type and the pollutant stock, thus allowing different types to suffer differently from pollution.

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, h(E_t, \theta_t)) \right]$$

I assume both U and h are continuously differentiable in all their arguments. Moreover, goods consumption provides utility ($U_c, U_q > 0$) whereas labor supply and pollution damages cause disutility ($U_l, U_E < 0$). Utility is concave in consumption of both goods ($U_{cc}, U_{qq} \leq 0$) and convex in labor and pollution damages ($U_{ll}, U_{hh} \geq 0$).

The differentiability hypothesis on the utility function is not innocuous[†], but nevertheless allows us to use the usual maximization methods. Aside from the presence of this consumption externality, the model is otherwise unchanged from the dynamic Mirrlees framework. Let us denote by θ_t the skill level at period t which is drawn from a set $\Theta \in \mathbb{R}_+$ and by $\theta^t \equiv \{\theta_0, \dots, \theta_t\}$ an agent's skill history up to period t which is drawn from the set $\Theta^t \in \mathbb{R}_+^t$. Agents start their life with a random skill level θ_0 drawn from a distribution $F_0(\theta)$, after which their skills follow a Markov process $F_t(\theta|\theta_{t-1})$ with probability density $f_t(\theta|\theta_{t-1})$.

[†]For example, utility functions requiring a minimum consumption requirement for the polluting good as in Klenert and Mattauch (2016) are not allowed here. This is an especially relevant case as low-income households are believed to spend a disproportionate amount in carbon-intensive goods.

Assumption 3.0.1. For all $t < T$, f_t is differentiable in both arguments with

$$f_{1,t} \equiv \frac{\partial f_t}{\partial \theta} \qquad f_{2,t} \equiv \frac{\partial t_t}{\partial \theta_{t-1}}$$

Moreover, for all θ_{t-1} ,

$$\phi_t(\theta|\theta_{t-1}) \equiv \frac{\theta_{t-1} \int_{\theta}^{\infty} f_{2,t}(x|\theta_{t-1}) dx}{\theta f_t(\theta|\theta_{t-1})} < 0 \quad \forall \theta \qquad \lim_{\theta \rightarrow \infty} \frac{1 - F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})} < \infty$$

As is standard in the literature, we abstract from the production side of the economy and therefore do not account for output pollution damages. Hence, labor income is simply a function of labor supply and type: $y_t = \theta_t l_t$. Let $c_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the clean good, $q_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the polluting good, and $y_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of effective labor. Finally, denote by $\sigma^t(\theta^t) : \Theta^t \rightarrow \Theta^t$ the agent's report in period t and by Σ^t the set of all possible reporting strategies in period t . The planner's problem is finding an allocation maximizing the social welfare function under incentive compatibility constraints and resource constraints as in Golosov, Kocherlakota, et al. (2003) while also taking into account the effect of current polluting good consumption on the pollutant stock.

$$\begin{aligned} & \max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), E_t\}_{\theta^t \in \Theta^t; t=0, \dots, T}} \int_0^{\infty} \alpha(\theta) \left(\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, h(E_t, \theta_t) \right) \middle| \theta \right] \right) dF_0(\theta) \\ & \text{s.t.} \\ & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, h(E_t, \theta_t) \right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, h(E_t, \theta_t) \right) \middle| \theta \right] \\ & \int_0^{\infty} \mathbb{E}_0 \left[\sum_{t=0}^T \frac{c_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta) \leq \int_0^{\infty} 0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta) \\ & E_t = S \left(E_0, E_1, \dots, E_{t-1}, \int_0^{\infty} q_t(\theta) dF_0(\theta) \right) \end{aligned}$$

Because we would like to compare our results to those in a static environment such as Cremer et al. (1998) and Jacobs and de Mooij (2015), we follow Golosov, Troshtkin, et al. (2016) who derive optimality conditions which resemble the classical results from the static Mirrlees literature (Mirrlees 1971; Mirrlees 1976; Saez 2001). and do not rely on weak separability like Golosov, Kocherlakota, et al. (2003). First off, we rewrite the incentive compatibility constraints in a recursive form. For all skill pairs $\hat{\theta}$ and θ and for all periods,

$$U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, h(E_t, \theta_t)\right) + \beta \omega_{t+1}(\theta^t | \theta_t) \geq U\left(c_t(\theta^{t-1}, \hat{\theta}), q_t(\theta^{t-1}, \hat{\theta}), \frac{y_t(\theta^{t-1}, \hat{\theta})}{\theta_t}, h(E_t, \theta_t)\right) + \beta \omega_{t+1}(\theta^t | \theta_t)$$

$$\omega_{t+1}(\theta^{t-1}, \hat{\theta} | \theta_t) = \mathbb{E}_t \left[\sum_{s=t+1}^T \beta^{s-t-1} U\left(c_s(\hat{\theta}^s, q_s(\hat{\theta}^s), \frac{y_s(\hat{\theta}^s)}{\theta_s}, h(E_s, \theta_s))\right) \middle| \theta_t \right]$$

Here, $\hat{\theta}^s$ denotes a history where a θ^s -agent misreports their type once. Next, the planner's problem is rewritten recursively.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, E_-) = \min_{c, q, y, u, w, w_2, E} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, h(E, \theta)) \right] f_t(\theta | \theta_-) d\theta$$

s.t.

$$\dot{u}(\theta) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, h(E, \theta) \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) f_t(\theta | \theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta$$

$$u(\theta) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, h(E, \theta) \right) + \beta w(\theta)$$

$$E = S \left(E_-, \int q(\theta) f(\theta | \theta_-) d\theta \right)$$

The only difference with the standard model is the presence of a polluting good which generates pollution that affects current and future utility.

4 Optimal wedges

Because θ enters l as well as h , we can write $U_\theta = -U_l \frac{l}{\theta} + U_h h_\theta$. Hence, after substituting U_θ , the Hamiltonian to the planner's problem is as follows.

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, E) \right] + \psi \left[-U_l(c, q, l, h(E, \theta)) \frac{l}{\theta} + U_h(c, q, l, h(E, \theta)) h_\theta + \beta w_2 \right] \\ & - \lambda_1 \bar{\alpha}_t u(\theta) f_t + \lambda_2 u(\theta) f_2 + \phi [u - U(c, q, l, E) - \beta w] - \mu \left[E - S \left(E_-, \int q(\theta) f(\theta | \theta_-) d\theta \right) \right] \end{aligned}$$

I derive the optimal labor wedge in the appendix.

Theorem 4.0.1. *The optimal labor wedge with pollution and heterogeneous type-dependent pollution damages are*

$$\frac{\tau^y}{1 - \tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

The labor wedge is identical to Golosov, Troshkin, et al. (2016) if $\xi^l = 0$ and $\xi^c = 0$, which happens if all agents are equally affected by pollution. Let's suppose $A_t(\theta) = A_{t-1}$ for simplicity. All things equal, $\xi^c \geq 0$ increases $C_t(\theta)$ and therefore increases the labor wedge. Intuitively, when higher pollution makes consumption more desirable, the planner is further prompted to redistribute income to ensure low-ability agents can consume more.

The savings wedge is defined as follows.

Theorem 4.0.2. *The optimal savings wedge with pollution and heterogeneous type-dependent pollution damages are*

$$\frac{1}{U_c(\theta^t)} - \frac{\psi}{\theta f}(\gamma - \xi^c) = \left(\frac{1}{\hat{\rho}_t} \frac{1}{\beta R} \right)^{T-t} \mathbb{E}_t \left[\frac{1}{U_c(\theta^T)} \right]$$

If $\psi \geq 0$, $\gamma \geq 0$, $\xi^c \geq 0$, and $\gamma \geq \xi^c$,

$$\frac{1}{U_c(\theta^t)} \geq \left(\frac{1}{\hat{\rho}_t} \frac{1}{\beta R} \right)^{T-t} \mathbb{E}_t \left[\frac{1}{U_c(\theta^T)} \right] \geq \left(\frac{1}{\hat{\rho}_t} \frac{1}{\beta R} \right)^{T-t} \frac{1}{\mathbb{E}_t[U_c(\theta^T)]}$$

This implies a positive savings wedge as in the standard model. However, the wedge is tampered by $\xi^c \geq 0$. Intuitively, if higher pollution makes consumption more desirable, there is a lesser need to discourage savings as agents will be tempted to consume now rather than save for later. If $\gamma \leq \xi^c$ however, this may even possibly call for a savings subsidy. The savings is identical to that in Golosov, Kocherlakota, et al. (2003) if there is full separability.

Theorem 4.0.3. *The optimal pollution wedge with heterogeneous type-dependent pollution damages is*

$$\tau^E = \frac{1}{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma - \kappa + \xi^c - \xi^q}{1 + \epsilon - \gamma + \xi^l - \xi^c}} \left[\left(-\frac{U_h}{U_c} h_E + \frac{1}{R} \frac{\partial V_{t+1}}{\partial E} \right) S_{2f} - \frac{U_h}{U_c} h_E \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1 + \xi^l + \xi^h - \gamma - \xi^c}{1 + \epsilon - \gamma + \xi^l - \xi^c} S_{2f} + \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma - \kappa + \xi^c - \xi^q}{1 + \epsilon - \gamma + \xi^l - \xi^c} \right]$$

The first term is the Pigouvian tax, the second term is a correction term applied to the MRS between pollution and consumption, and the last term only depends on elasticities. Individualized pollution damages do not affect anything here if $\xi^c = \xi^q$ and $\xi^l = \xi^c$. In words, clean good consumption and polluting good consumption expose one to equal environmental harm. Moreover, supplying labor exposes one to as much environmental harm as consuming clean goods. In the first case, there is no sense encouraging or discouraging polluting good consumption because it does not protect against pollution damages. In the second case, a greater positive ξ^l would mean hard-workers are more exposed to pollution damages and the planner would therefore like to discourage the polluting good less (???). Similarly, a greater positive ξ^c would mean those who consume more clean goods are more exposed to pollution damages and the planner would therefore like to discourage the polluting good more (???).

4.1 Pollution wedge

We can write the pollution wedge as follows.

$$\tau^E(\theta_-) = \frac{1}{1 + \left(\frac{\tau^y}{1 - \tau^y}\right) \frac{\kappa - \gamma}{1 + \epsilon - \gamma}} \left[\int_0^\infty \left(-\frac{U_E}{U_c} + \frac{1}{R} \frac{\partial V_{t+1}}{\partial E} f \right) S_2 f(\theta|\theta_-) d\theta - \int_0^\infty \frac{U_E}{U_c} \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi - \gamma}{1 + \epsilon - \gamma} S_2 f(\theta|\theta_-) d\theta - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\kappa - \gamma}{1 + \epsilon - \gamma} \right]$$

There are three distinct terms in parentheses. The first term is the Pigouvian correction term equal to the discounted sum of the marginal damages caused by an extra unit of emissions today and tomorrow, the second term is a fiscal correction term, and the third term is the pure commodity taxation component. These two last terms equal zero if the labor wedge is zero (such as when there are no inequality concerns) or if $\gamma = \kappa = \xi = 0$ (such as when utility is weakly separable in all arguments).

The commodity taxation component is only non-zero if $\gamma \neq \kappa$. In other words, the only reason the planner would like to encourage or discourage consumption of the polluting good is if it was more or less complementary with labor supply. This result translates directly from the static Mirrlees literature (Cremer et al. 1998; Jacobs and de Mooij 2015).

Assuming that pollution affects utility, the fiscal correction term is only non-zero if $\gamma \neq \xi$. Otherwise, this means pollution is just as labor-boosting (labor-dampening) as clean good consumption and there is therefore no need to deviate from the Pigouvian tax. Again, this result already exists in the static Mirrlees literature.

Unlike the static Mirrlees case however, the integrals are conditional on past skill shock history. This means that each type faces a different pollution wedge. This makes sense because different types may have different consumption profiles such that the expected MRS between consumption and the environment will differ. To fix ideas, consider the fully separable case.

$$\tau^E(\theta_-) = \int_0^\infty \left(-\frac{U_E}{U_c} + \frac{1}{R} \frac{\partial V_{t+1}}{\partial E} f \right) S_2 f(\theta|\theta_-) d\theta$$

We know U_E is the same for everyone, so only U_c will affect the optimal pollution wedge. If we expect consumption to be high, U_c becomes smaller such that $\frac{U_E}{U_c}$ becomes bigger. In other words, high-ability agents should pay a higher price on pollution than low-ability agents.

5 Quantitative exercise

6 Conclusion

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A Derivation of optimal wedges

A.1 Labor wedges

The first-order conditions are as follows.

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta &= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} + \psi U_{qh} h_\theta + \mu S_q f &= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l &= \frac{1}{\theta} \psi [U_{ll} l + U_l] - \psi U_{hl} h_\theta \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 \bar{\rho}_t f + \lambda_2 f_2 + \phi &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial E} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial E} f - \psi U_{lh} h_E \frac{l}{\theta} + \psi U_{hh} h_E h_\theta + \psi U_h h_{\theta E} - \psi U_h h_E = \mu
\end{aligned}$$

If we substitute μ using $\frac{\partial \mathcal{H}}{\partial E}$ and ϕ using $\frac{\partial \mathcal{H}}{\partial c}$,

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} + \psi U_{qh} h_\theta + \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial E} f - \psi U_{lh} h_E \frac{l}{\theta} + \psi U_{hh} h_E h_\theta + \psi U_h h_{\theta E} - \frac{f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta}{U_c} U_h h_E \right) S_2 f = \frac{f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta}{U_c} U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \frac{f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta}{U_c} U_l &= \frac{1}{\theta} \psi [U_{ll} l + U_l] - \psi U_{hl} h_\theta \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 \bar{\rho}_t f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta}{U_c} &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \frac{f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta}{U_c} \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta
\end{aligned}$$

Suppose $h(E, \theta) = \frac{E}{\theta}$ to simplify derivations. In that case,

$$h_\theta = -\frac{E}{\theta^2} \qquad h_E = \frac{1}{\theta} \qquad h_{\theta E} = -\frac{1}{\theta^2}$$

So the first-order conditions become

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial q} : f - \psi U_{ql} \frac{l}{\theta} - \psi U_{qh} \frac{E}{\theta^2} + \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial E} f - \psi U_{lh} \frac{l}{\theta^2} - \psi U_{hh} \frac{h}{\theta^2} - \psi U_h \frac{1}{\theta^2} - \frac{f - \psi U_{cl} \frac{l}{\theta} - \psi U_{ch} \frac{E}{\theta^2}}{U_c} U_h \frac{1}{\theta} \right) S_2 f &= \frac{f - \psi U_{cl} \frac{l}{\theta} - \psi U_{ch} \frac{E}{\theta^2}}{U_c} U_q \\ \frac{\partial \mathcal{H}}{\partial l} : -\theta f - \frac{f - \psi U_{cl} \frac{l}{\theta} - \psi U_{ch} \frac{E}{\theta^2}}{U_c} U_l &= \frac{1}{\theta} \psi [U_{ll} l + U_l] + \psi U_{hl} \frac{E}{\theta^2} \\ \frac{\partial \mathcal{H}}{\partial u} : -\lambda_1 \bar{\rho}_t f + \lambda_2 f_2 + \phi &= \dot{\psi} \\ \frac{\partial \mathcal{H}}{\partial w} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \frac{f - \psi U_{cl} \frac{l}{\theta} - \psi U_{ch} \frac{E}{\theta^2}}{U_c} \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \end{aligned}$$

Using $\epsilon_t(\theta) \equiv \frac{U_{ll,t}(\theta) l_t(\theta)}{U_{l,t}(\theta)}$ and $\gamma_t(\theta) \equiv \frac{U_{cl,t}(\theta) l_t(\theta)}{U_{c,t}(\theta)}$, we can rewrite $\frac{\partial \mathcal{H}}{\partial l}$. By $h = E/\theta$, we can also write $h_\theta = -h/\theta$. Let $\xi_t^l(\theta) \equiv \frac{U_{hl,t}(\theta) h_t(\theta)}{U_{l,t}(\theta)}$ and $\xi_t^c(\theta) \equiv \frac{U_{ch,t}(\theta) h_t(\theta)}{U_{c,t}(\theta)}$.

$$\left(\frac{U_l}{\theta U_c} + 1 \right) \theta f = \frac{1}{\theta} \psi (1 + \epsilon - \gamma + \xi^l - \xi^c) (-U_l)$$

Since $\tau^y = 1 + \frac{U_l}{\theta U_c}$,

$$\frac{\tau^y}{1 - \tau^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon - \gamma + \xi^l - \xi^c)$$

$$\psi = -\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} f$$

Likewise, we can find another expression for ψ by isolating it in $\frac{\partial \mathcal{H}}{\partial l}$.

$$\frac{\tau^y}{1 - \tau^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon - \gamma + \xi^l - \xi^c) \implies \psi = \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\theta f}{U_c} \frac{1}{1 + \epsilon - \gamma + \xi^l - \xi^c}$$

Bringing the two together,

$$-\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} f = \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\theta f}{U_c} \frac{1}{1 + \epsilon - \gamma + \xi^l - \xi^c} \implies -\frac{\partial V_{t+1}}{\partial w_2} = \beta R \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\theta}{U_c} \frac{1}{1 + \epsilon - \gamma + \xi^l - \xi^c}$$

By the envelope condition, $\frac{\partial V_t}{\partial \bar{w}} = \lambda_1$ and $\frac{\partial V_t}{\partial \bar{w}_2} = -\lambda_2$. Hence,

$$\lambda_{2,t+1} = \beta R \left(\frac{\tau_t^y}{1 - \tau_t^y} \right) \frac{\theta}{U_{c,t}} \frac{1}{1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t(\theta)^l - \xi_t(\theta)^c}$$

To find the expression for $\psi(\theta)$, we can integrate $\frac{\partial \mathcal{H}}{\partial u}$ from θ to ∞ . This can be done by first noting that this is a first order differential equation of the form $\dot{\psi} = Q + P\psi$, where P is the first three terms of the rewritten first order condition with respect to u .

$$\begin{aligned} \dot{\psi} &= -\lambda_1 \bar{\rho}_t f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta} + \psi U_{ch} h_\theta}{U_c} \\ &= -\lambda_1 \bar{\rho}_t f + \lambda_2 f_2 + \frac{f}{U_c} - \frac{1}{\theta} (\gamma - \xi^c) \psi \end{aligned}$$

After solving the above, we should find

$$\psi(\theta) = \int_\theta^\infty \exp \left[- \int_\theta^x (\gamma(\tilde{x}) - \xi^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_c(x)} f(x) - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

We can isolate λ_1 using the boundary condition $\psi(0) = 0$.

$$\lambda_{1,t} = \frac{\int_0^\infty \exp \left[- \int_0^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^\infty \exp \left[- \int_0^x (\gamma(\tilde{x}) - \xi^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] \bar{\alpha}_t(x) f_t(x) dx}$$

Now, we substitute $\psi(\theta)$ in the expression for the labor wedge.

$$\begin{aligned} \frac{\tau^y}{1 - \tau^y} &= \psi U_c \theta f(1 + \epsilon - \gamma + \xi^l - \xi^c) \\ &= \int_\theta^\infty \exp \left[- \int_\theta^x (\gamma(\tilde{x}) - \xi^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_c(x)} f(x) - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx \frac{U_c}{\theta f} (1 + \epsilon - \gamma + \xi^l - \xi^c) \end{aligned}$$

If we substitute λ_2 , iterating $\lambda_{2,t+1}$ back to get $\lambda_{2,t}$,

$$\begin{aligned} \frac{\tau^y}{1 - \tau^y} &= \int_\theta^\infty \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_\theta^x (\gamma(\tilde{x}) - \xi^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] (1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)) \frac{f_t(x)}{\theta_t f_t(\theta)} dx - \\ &\quad \int_\theta^\infty \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_\theta^x (\gamma(\tilde{x}) - \xi^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] U_{c,t}(x) (1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)) \lambda_1 \bar{\alpha}_t(x) \frac{f_t(x)}{\theta_t f_t(\theta)} dx - \\ &\quad \beta R \left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) \frac{1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)}{1 + \epsilon_{t-1}(\theta) - \gamma_{t-1}(\theta) + \xi_{t-1}^l(\theta) - \xi_{t-1}^c(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_\theta^\infty \exp \left[- \int_\theta^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} \end{aligned}$$

We can rewrite the common terms in the first two lines as follows.

$$\begin{aligned} \frac{U_{c,t}(\theta_t)}{U_{c,t}(x)} \exp \left[- \int_\theta^x (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] &= \exp \left[\ln \left(\frac{U_{c,t}(\theta_t)}{U_{c,t}(x)} \right) - \int_\theta^x (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] \\ &= \exp \left[- \int_\theta^x \frac{U_{c\theta,t}(\tilde{x})}{U_{c,t}(\tilde{x})} d\tilde{x} - \int_\theta^x (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{d\tilde{x}}{\tilde{x}} \right] \\ &= \exp \left[\int_\theta^x \sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} d\tilde{x} - \int_\theta^x (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} d\tilde{x} \right] \\ &= \exp \left[\int_\theta^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] \end{aligned}$$

From here, we can write the optimal labor wedge in terms of elasticities.

$$\frac{\tau^y}{1 - \tau^y} = \beta R \left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) \frac{1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)}{1 + \epsilon_{t-1}(\theta) - \gamma_{t-1}(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)}$$

This gives us the familiar ABCD form from Golosov, Troshkin, et al. (2016), with the exception of the presence of ξ^l and ξ^c .

$$\frac{\tau^y}{1 - \tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_t(\theta) + \xi_t^l(\theta) - \xi_t^c(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - (\gamma_t(\tilde{x}) - \xi_t^c(\tilde{x})) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

A.2 Capital wedges

TBA

A.3 Pollution wedges

First, we can rewrite $\frac{\partial \mathcal{H}}{\partial q}$ using $1 + \tau^E = \frac{U_q}{U_c}$.

$$\tau^E = \frac{U_E}{U_c} S_q f + \frac{\left(1 + \left[\frac{1}{R} \frac{\partial V_{t+1}}{\partial E} f - \psi U_{lE} \frac{l}{\theta}\right] S_q\right) f - \psi U_{ql} \frac{l}{\theta}}{f - \psi U_{cl} \frac{l}{\theta}} - 1$$

Next, we substitute the ψ we found in the appendix on labor wedges and express everything in term of elasticities.

$$\tau^E = -\frac{U_E}{U_c} S_q f + \frac{1 + \left[\frac{1}{R} \frac{\partial V_{t+1}}{\partial E} - \left(\frac{\tau^y}{1-\tau^y}\right) \frac{\xi}{1+\epsilon-\gamma} \frac{U_E}{U_c}\right] S_q f - \left(\frac{\tau^y}{1-\tau^y}\right) \frac{\kappa}{1+\epsilon-\gamma} \frac{U_q}{U_c}}{1 - \left(\frac{\tau^y}{1-\tau^y}\right) \frac{\gamma}{1+\epsilon-\gamma}} - 1$$

Or, rewritten a bit more simply,

$$\tau^E = \frac{1}{1 - \left(\frac{\tau^y}{1-\tau^y}\right) \frac{\gamma-\kappa}{1+\epsilon-\gamma}} \left[\left(-\frac{U_E}{U_c} + \frac{1}{R} \frac{\partial V_{t+1}}{\partial E}\right) S_q f - \frac{U_E}{U_c} \left(\frac{\tau^y}{1-\tau^y}\right) \frac{\xi-\gamma}{1+\epsilon-\gamma} S_q f + \left(\frac{\tau^y}{1-\tau^y}\right) \frac{\gamma-\kappa}{1+\epsilon-\gamma} \right]$$